# Literature status: Schlumprecht's scalar-plus-compact question

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## Status

This is a literature-already-answered packet, not an original proof. It records that Question Q1 from Th. Schlumprecht, *How many operators do there exist on a Banach space?*, arXiv:math/0208055, is answered by S. A. Argyros and R. G. Haydon, *A hereditarily indecomposable $\mathscr{L}_\infty$-space that solves the scalar-plus-compact problem*, arXiv:0903.3921.

## Source question

In item (Q1) of the introduction of arXiv:math/0208055, Schlumprecht asks:

> Assume $X$ is an infinite dimensional Banach space. Does there exist a linear bounded operator $T$ on $X$ which is not of the form $T = \lambda I + C$, where $\lambda$ is a scalar and $C$ is a compact operator?

The same introduction explains that a negative answer would be a Banach space on which every operator is a scalar multiple of the identity plus a compact operator.

## Supporting answer

Argyros and Haydon's arXiv:0903.3921 is explicitly titled as solving the scalar-plus-compact problem. Its abstract states that they construct a hereditarily indecomposable Banach space with dual isomorphic to $\ell_1$ and that every bounded linear operator on this space is expressible as $\lambda I + K$, with $K$ compact.

The introduction says that the paper solves the scalar-plus-compact problem. The final theorem of the section titled *Operators* states: if $T$ is a bounded linear operator on the constructed space $\mathfrak{X}_K$, then there is a scalar $\lambda$ such that $T - \lambda I$ is compact.

Therefore Schlumprecht's Q1 has a negative answer in the later literature.

## Scope

This packet does not claim any answer to Schlumprecht's Q2/Q3 subspace version, Q4 Class 1/Class 2 dichotomy, Q5 partial-unconditionality question, or the remark asking for an elementary proof that a Class 1 space contains an unconditional basic sequence. Those are separate targets.